

Symbolic Regression for Precision LHC Physics

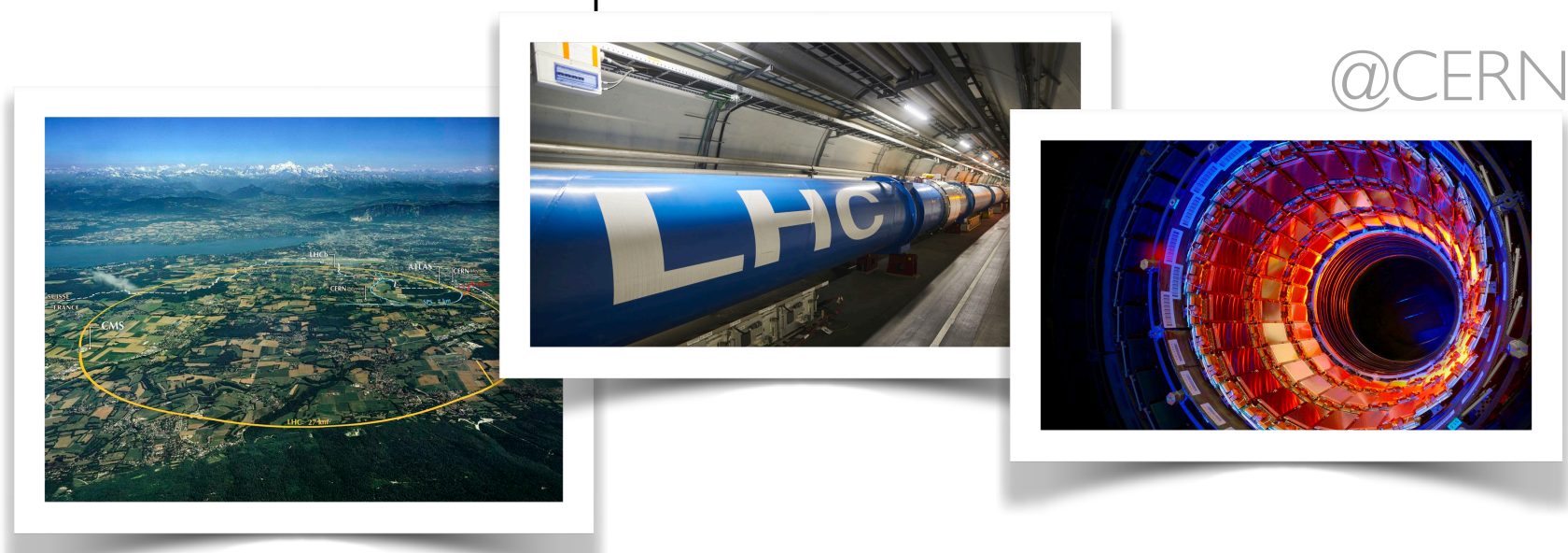
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Introduction

- In high energy physics (HEP), fundamental interactions can be described by the **Standard Model** (SM) of particle physics to a good extent.
- The SM is tested at the world's most powerful microscope: the **Large Hadron Collider** (LHC). Theoretical predictions and observables are compared.



- Formulas** are the actual language of theoretical predictions [1].
- The question:** Is there a way to find simple, accurate, and closed analytical formulas from noisy, potentially high dimensional datasets?

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Methodology

- We use **symbolic regression** (SR) with the PySR [2] library.
- Formulas are described as expression trees, as shown in Fig. 1. An evolutionary algorithm implements mutations/crossover between trees, as shown in Fig. 2, and new trees arise.

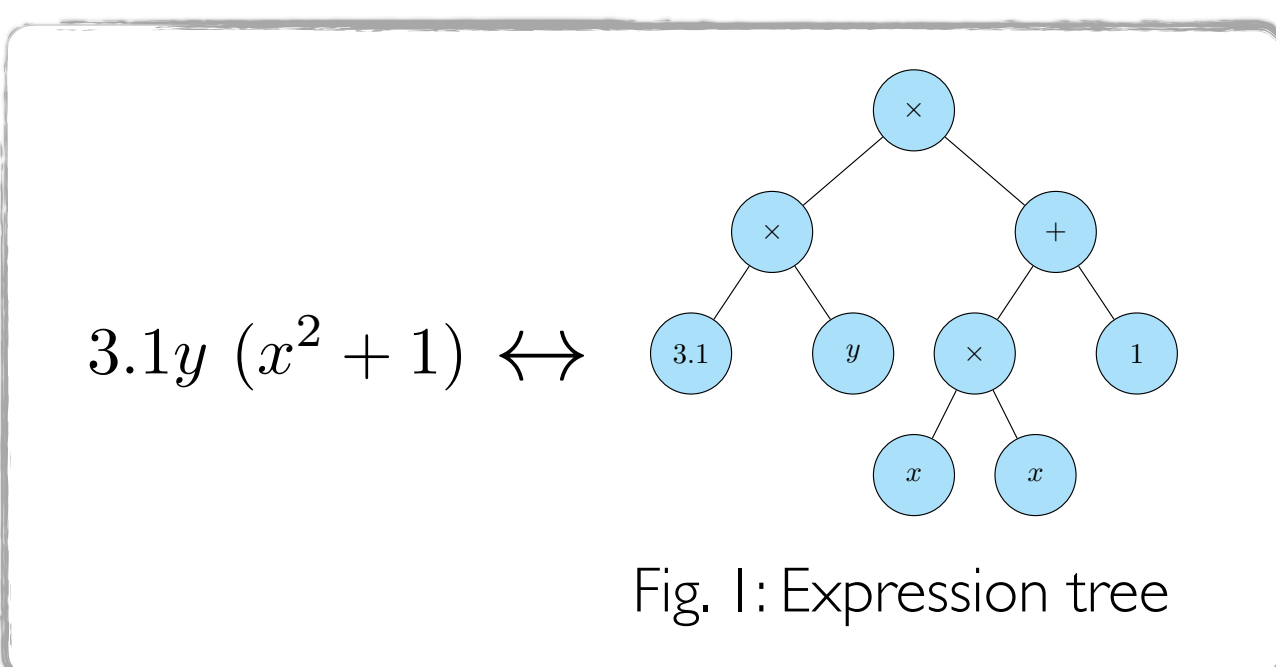


Fig. 1: Expression tree

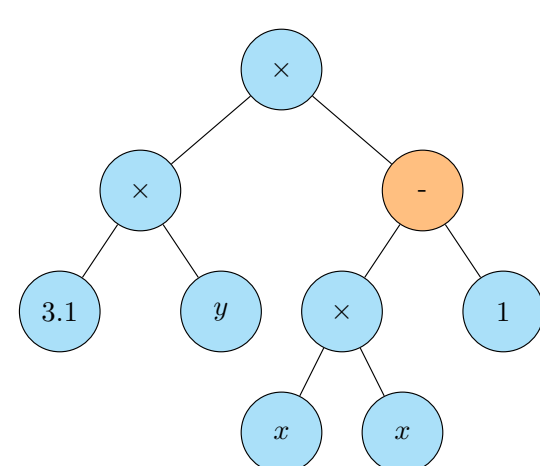


Fig. 2: Mutation

- Each tree has a complexity c , related to its number of nodes.
- Fittest trees optimise a certain metric

Accuracy:	Score:	Best:
Minimise	Maximise	Highest score with
$L = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$	$-\frac{\partial \log(L)}{\partial c}$	$L \leq 1.5 \times L_{\min}$

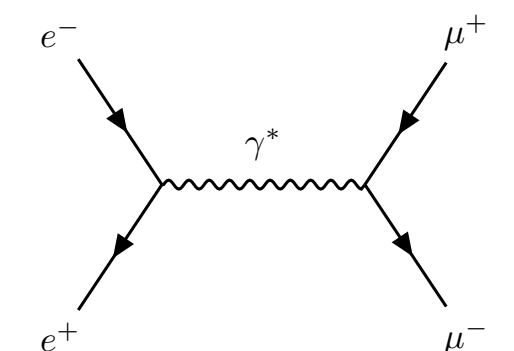
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Results

- It is important that SR can perform **equation recovery**. Known laws are essential benchmarks.

- Example

$$e^- e^+ \rightarrow \mu^- \mu^+$$



- We obtain the distributions of kinematic variables (cosines of momentum of the outgoing leptons, in this case) from simulations and train a regressor on them. Different binnings are used to assess robustness against statistical uncertainty.

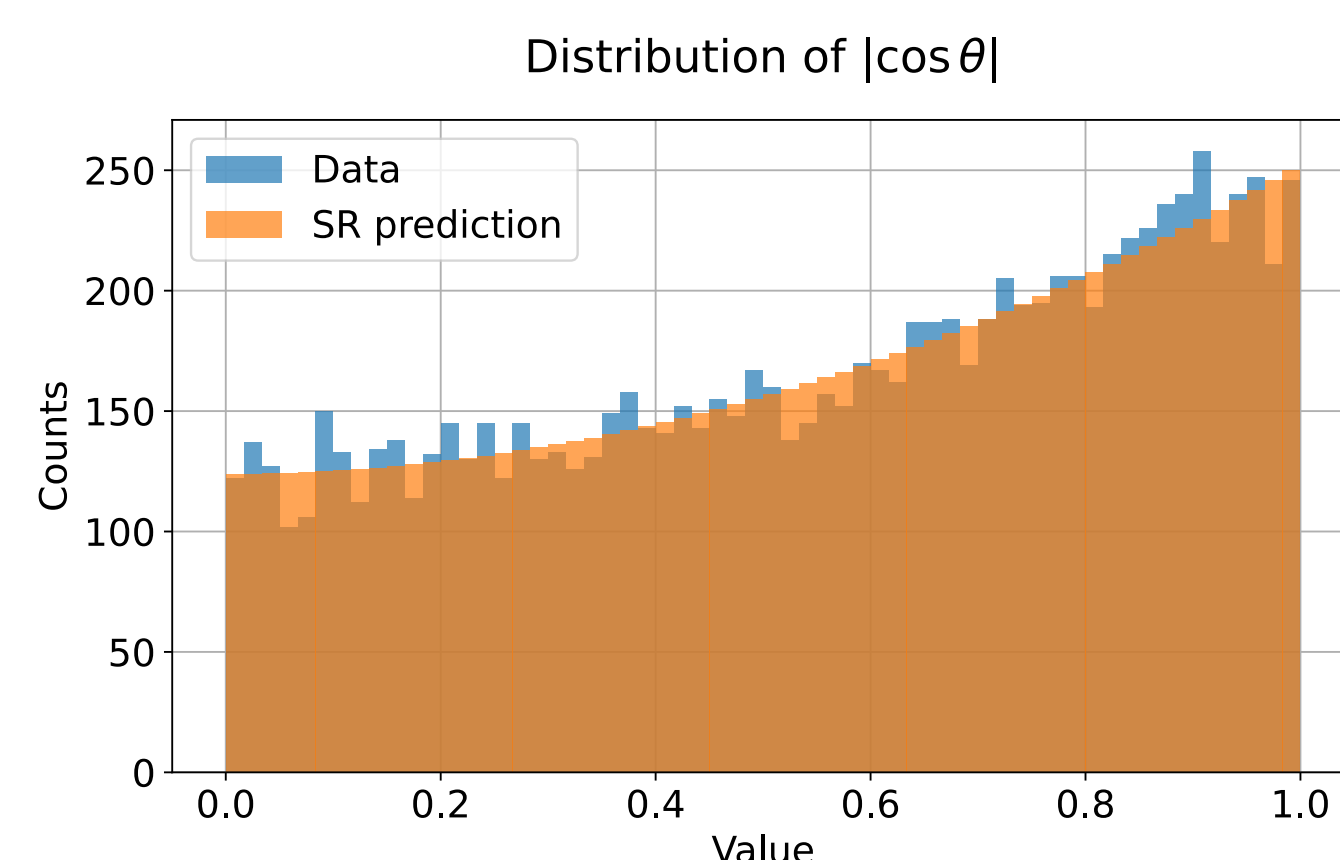


Fig. 3: Data and SR prediction.

Bins	Accuracy	Score	Best
10	x_0^2 (296.52358194355 $x_0^4 + 7046.0674$) + 7613.42	7250.1396 · x_0^2 + 7589.319	7250.1396 · x_0^2 + 7589.319
30	x_0^2 (123.43398 x_0^4 2326.98053420264)+ 2538.3494	2415.3643 x_0 2125.6453	2417.7627 x_0^2 + 2527.635
100	x_0 (207.340216 x_0 + 428.81232) 109.830989048 + 750.30175	725.2477 x_0 637.3749	726.08685 x_0^2 + 757.9762

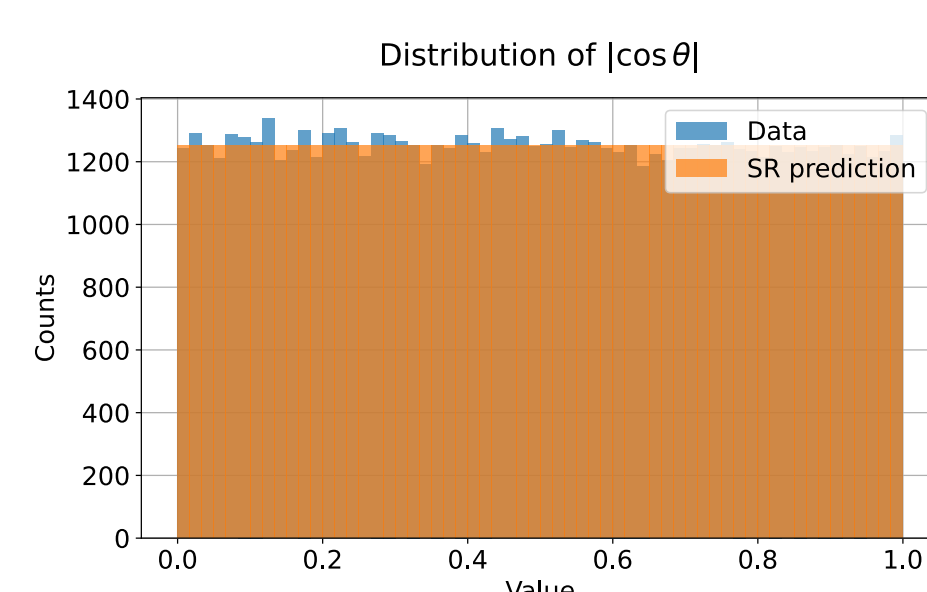
Tab. 1: SR formulas.

$$x_0 = \cos \theta$$

- Success in all cases when compared to analytical QFT differential distribution in 'Best' criterion, as shown in Fig. 3 and Tab. 1.

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4s} (1 + \cos^2 \theta)$$

- Gaussian denoising can be used to recover invariants in reweighted distributions.



Denoise	Accuracy	Score	Best
True	6833.678	6833.678	6833.678
False	x_0^2 (661.2393 x_0 706.5968)(x_0^2 + x_0) + 7593.010	7614.0547 194.02724 x_0	7614.049 194.01741 x_0

$$\sigma = \frac{4\pi\alpha^2}{3s}$$

- Now, working in SR angular coefficients [3] of $p p \rightarrow \gamma^*/Z \rightarrow l^+ l^-$

$$\frac{d^5\sigma}{dp_T d\eta dm d\cos\theta d\phi} = \frac{3}{16\pi} \frac{d^3\sigma^{U+L}}{dp_T d\eta dm} \left[(1 + \cos^2 \theta) + \sum_{i=0}^7 P_i(\theta, \phi) A_i \right]$$

- [1] Weinberg, S. *The Quantum Theory of Fields* (1996).
- [2] Cranmer, M. Interpretable Machine Learning with PySR and SymbolicRegression.jl, arXiv 2305.01582.
- [3] ATLAS Collaboration, arXiv 2309.09318.